

Time Complexity deletion at end = $O(n)$

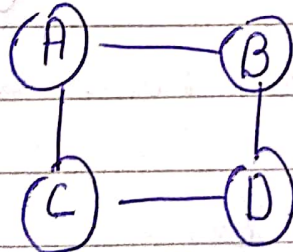
Graphs

A Graph G consists of two things.

- (I) A set V of element called nodes (or points or vertices).
- (II) A set E of edge such that each edge e in E is identified with a unique (unordered) pair $[u, v]$ of nodes in V , denoted by $e = [u, v]$

$$G = (V, E)$$

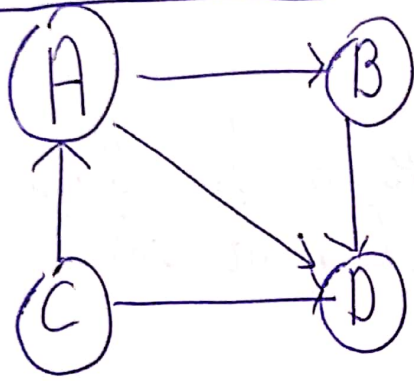
Adjacent nodes / Neighbours



A is adjacent to B, C
B is adjacent to A, D
C is adjacent to D, A
D is adjacent to B, C

The degree of a node u is written $\deg(u)$ is

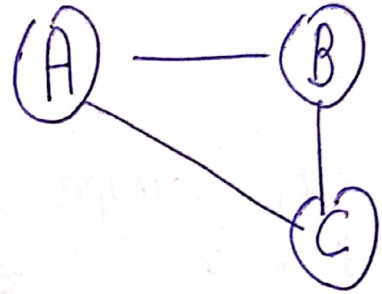
the no. of edges containing u .
 $\deg(u) = 0$ is called isolated node



Directed graph

In degree $(A) = 1$ (Incoming)

Out degree $(A) = 2$ (Outgoing)



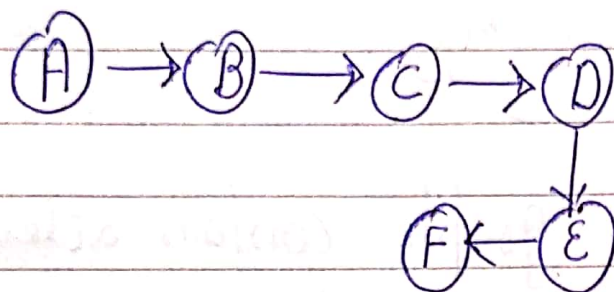
Undirected graph

A path is a sequence of vertices where each adjacent pair is connected by an edge. A path of length n from a node u to a node v is defined as a sequence of $n+1$ nodes.

$$P = (v_0, v_1, v_2, \dots, v_n)$$

Such that $u = v_0$; v_{i-1} is adjacent to v_i for

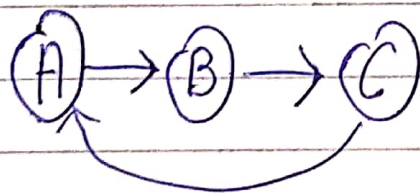
$i = 1, 2, 3, \dots, n$ and $v_n = v$.



Types of path:-

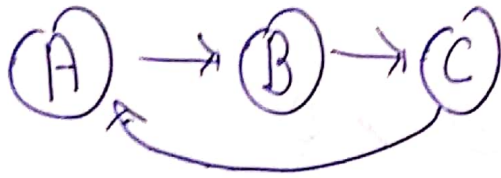
(I) Simple path - A simple path where all vertices are distinct, meaning no vertices is along the way.

(II) Closed path - (or cycle) - A closed path that start and end at the same vertex, forming a cycle. All other vertices are distinct.

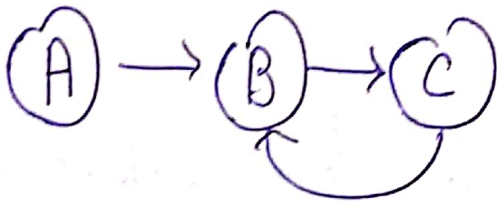


- A cycle is closed simple path with length 3 or more. A cycle of length k is called a k -cycle.

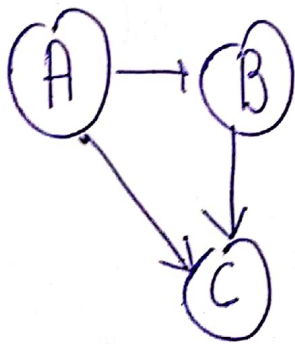
Cycle



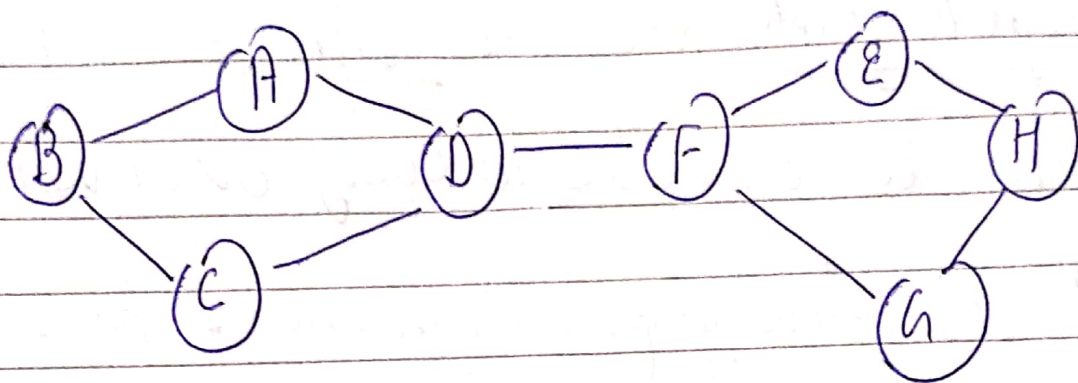
Cyclic graph - graph contain atleast one cycle



Acyclic graph - An acyclic graph is a graph that has no cycle. It may be directed or Undirected



Connected graph → A graph is said to be connected if there is a path b/w every pair of vertex.



Undirected Connected graph

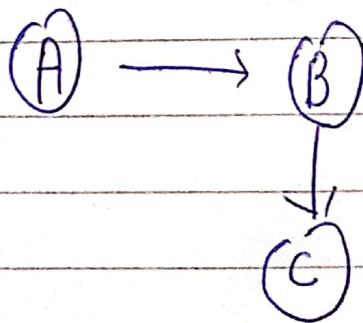
~~Complete~~

Directed Connected graph are two types:-

(I) Strongly Connected graph:- A directed graph is strongly connected if, for every pair of vertices u and v , there is a directed path from u to v and v to u .

(II) Weakly connected graph:- A undirected graph is weakly connected if when ignoring the direction of edges, the underlying graph undirected graph is connected or not

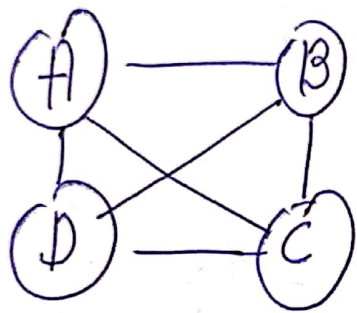
~~Complete~~ g



Not connected graph

Complete graph is a graph in which each ~~edge~~ vertex is connected to every other vertex.

A complete graph with n nodes have $\frac{n(n-1)}{2}$ edges.

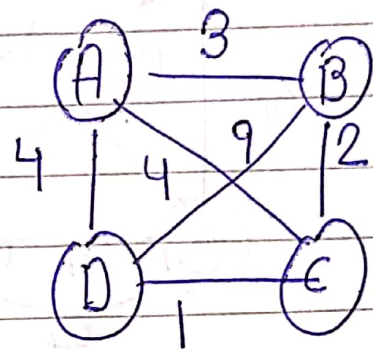
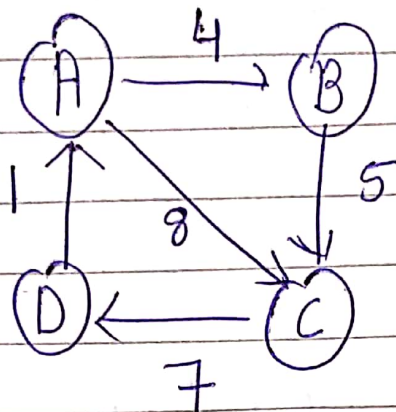


A connected graph w/o any cycle is called a tree graph or free tree or tree. This

means in particular, that there is a unique simple path P b/w any two nodes u and v in T .

Furthermore if T is a finite tree with n nodes, then T will have $n-1$ edges.

Weighted graph- A is said to be weighted graph if each edge in G is assigned a non-negative numerical value $w(e)$ is called the weight or length of e .

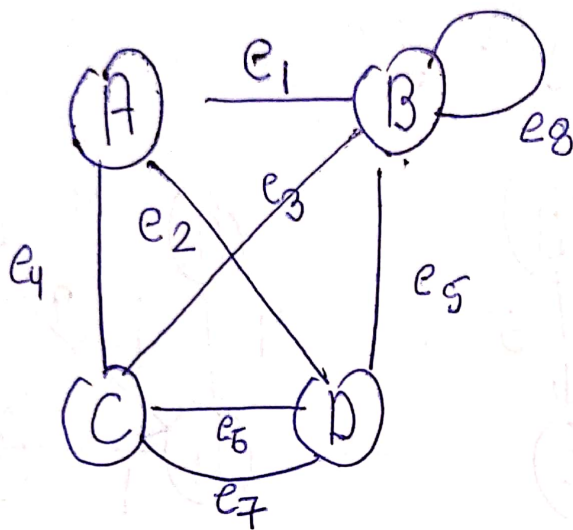


(I) Multiple edge - Distinct edges e and e' are called multiple edges if they connect the same endpoint, that is if $e = [u, v]$ and $e' = [u, v]$

(II) ~~A~~ loops:- An edge e is called a loop if it has identical end points, i.e. if $e = [u, u]$

~~A~~ Multigraph - A graph with multiple edges and/or loop is called a multigraph.

A graph does not allow either multiple edges or loops.



This is multigraph but not a graph.

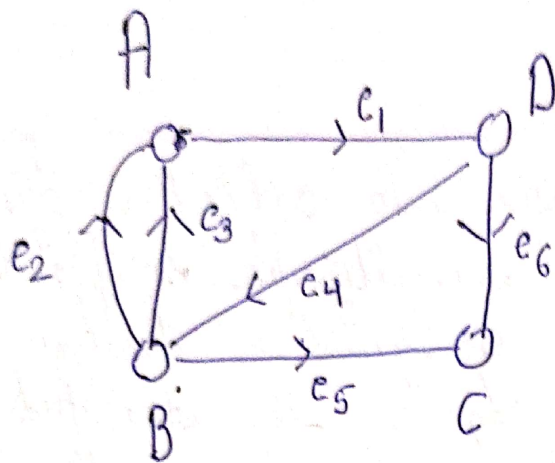
Unilaterally connected graph - A unilaterally connected graph ensures that there is at least one path b/w any two vertices, either going one way to other, but does not guarantee a bidirectional path like a strongly connected graph.

A directed graph G are also called a digraph or graph, is the same as multigraph except that each edge e in G is assigned a direction, or in other words, each edge e is identified with an ordered pair (u, v) of nodes in G rather than an unordered pair $[u, v]$.

A node u is called a source if it has a positive outdegree but zero indegree.

Also, u is called a sink if it has a zero outdegree but a positive indegree.

A directed graph G is said to be connected or strongly connected if for each pair of u, v of nodes in G there is a path from u to v and there is also a path from v to u . On the other hand G is said to be unilaterally connected if for any pair u, v of nodes in G there is a path from u to v or a path from v to u .



$P_1 = (D C B A)$ is not a path
 $P_2 = (D B A)$ is a path

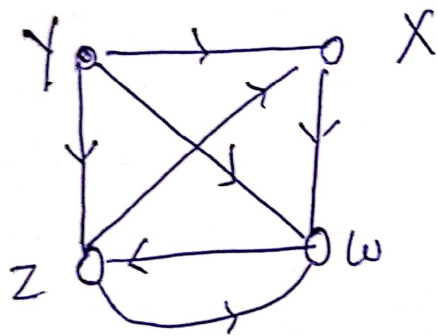
Graph Representation

Adjacency matrix

$$a_{ij} = \begin{cases} 1 & \text{if } v_i \text{ is adjacent to } v_j, \text{ that is, there is an edge} \\ 0 & \text{otherwise} \end{cases}$$

Suppose G is a simple directed graph with n nodes and suppose the nodes have been ordered and are called $v_1, v_2, \dots, v_n$. Then adjacency matrix $A = (a_{ij})$ of the graph G is the $n \times n$ matrix defined as follows

Such a matrix A , which contains entries of only 0 and 1 is called bit matrix or a boolean matrix

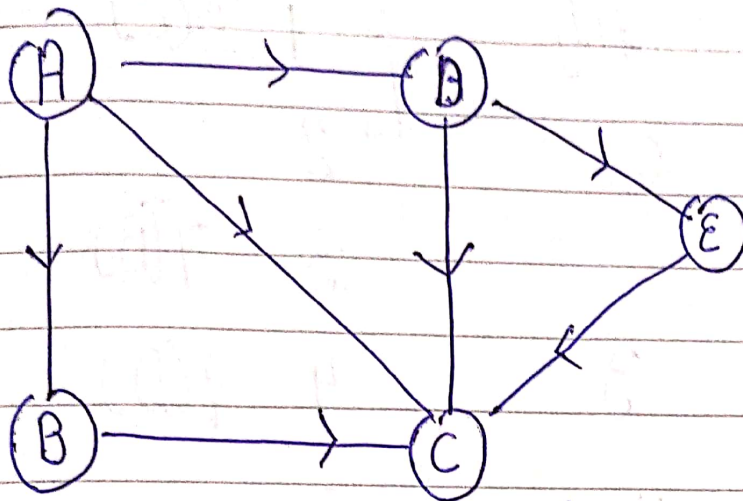


$$A = \begin{matrix} & \begin{matrix} X & Y & W & Z \end{matrix} \\ \begin{matrix} X \\ Y \\ W \\ Z \end{matrix} & \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

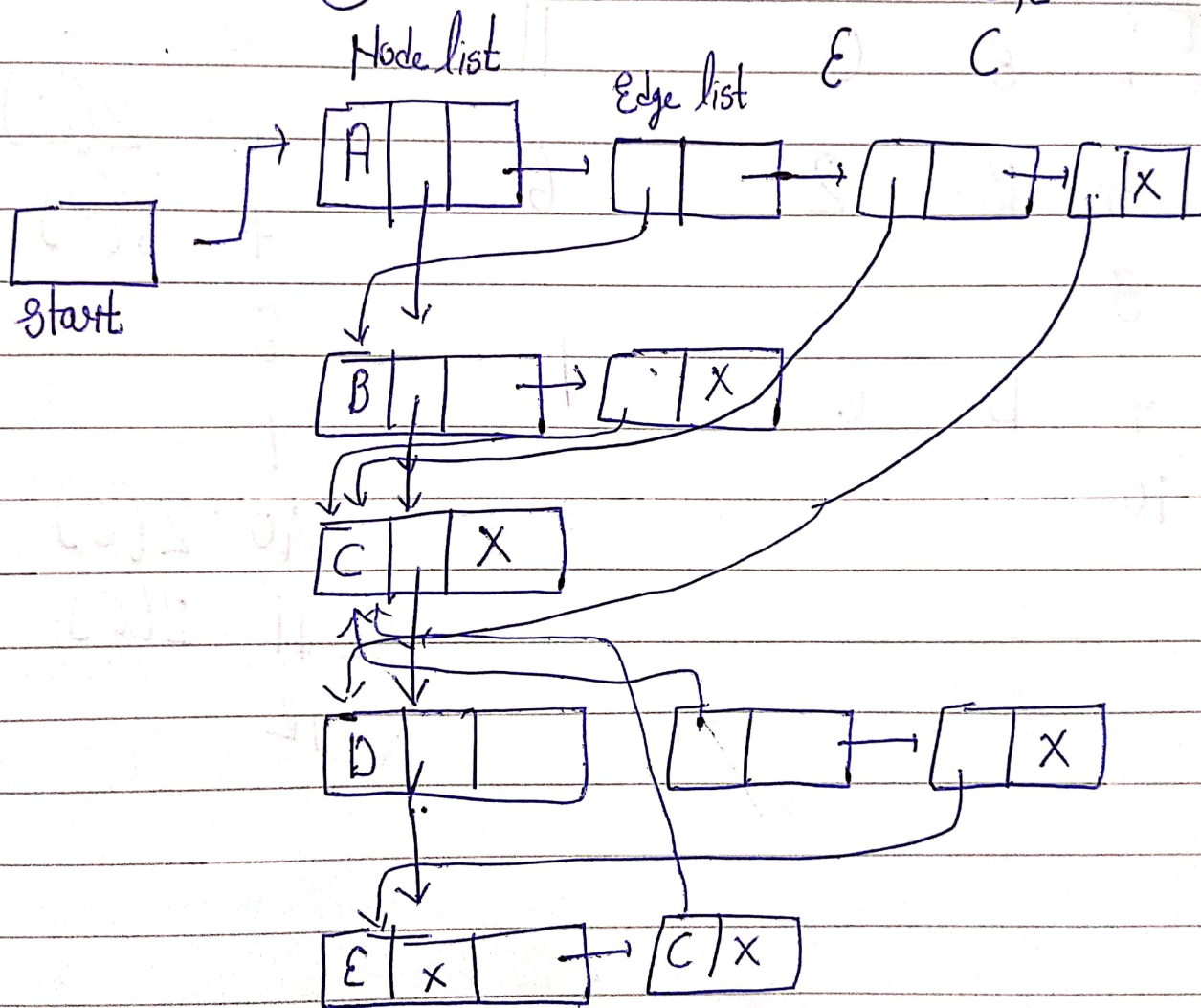
Adjacency list

If the graph is sparse graph then adjacency list is better.

linked list representation / adjacency structure



Node	Adjacency list
A	B, C, D
B	C
C	D, E
D	E
E	



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